

Booklet



**Dielectric Data Plotting and
Analyzing Tool
&
Polarization Data Plotting and
Analyzing Tool**



Developed

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Introduction

Generally, in most of the advanced functional Material labs, electrical properties of dielectric materials are investigated by the temperature/frequency dependent dielectric and electric field stimulated polarization measurements. Dielectric measurement is carried out by the Novocontrol and Waynekerr impedance analyzer. Polarization measurement is done by using TF analyzer, aixACCT system. These instruments generate a massive number of data in text file (Novocontrol) or excel sheet (Waynekerr) and a DAT file (aixACCT system) for respective measurements. These massive data needs to go through under rigorous shortening process to make them in proper format to further analysis and plotting of the various parameters or properties. In view of this problem, “Dielectric Data Plotting and Analyzing Tool (DDPAT)” and “Polarization Data Plotting and Analyzing Tool (PDPAT)” for dielectric data and polarization data respectively have been developed by the researchers of “Advanced Functional Materials Laboratory” (AFML), Department of Physics, IIT Hyderabad through Python program to facilitate the other researchers to precise analyzing and plotting the data among various parameters. Importance and procedure of using these tools are discussed in following topics:

- (1) Dielectric Data Plotting and Analyzing Tool (DDPAT)
 - (2) Polarization Data Plotting and Analyzing Tool(PDPAT)
-



Tool [1]

Dielectric Data Plotting and Analyzing Tool (DDPAT)

Dielectric data comprises frequency and temperature-dependent fundamental impedance parameters $C(F)$, $\tan(\delta)$, $\text{mod } Z(\text{ohm})$ which are independent of the geometry of the sample. Using these parameters, additional and normalized (by using the geometry of sample) impedance parameters such as ϵ' , ϵ'' , $\text{mod } \epsilon$, $\text{Real } Z$, $\text{Img } Z$, $\text{Real } M$, $\text{Img } M$, $\text{mod } M$, AC conductivity, $\text{Real } Y$, $\text{Img } Y$, $\text{Mod } Y$ etc. can be calculated by mathematical formulas. Plotting and analyzing these parameters in the Origin software at all or a specific range of temperatures (T) or frequencies (f) is either an arduous or time-consuming task for a researcher. It has two main reasons (1). it requires the calculation of each impedance parameter (ϵ' , ϵ'' , $\text{mod } \epsilon$, $\text{Real } Z$, $\text{Img } Z$, $\text{Real } M$, $\text{Img } M$, $\text{mod } M$, AC conductivity, $\text{Real } Y$, $\text{Img } Y$, $\text{Mod } Y$ etc.) separately along with normalization by the geometry of the sample using a mathematical formula. (2). It needs sorting the parameters with respect to temperatures/frequencies to plot in the Origin software. Sometimes, it seems impossible to plot the data between specific two rarely used impedance parameters (eg. $\text{Imp } Y$ Vs $\text{Img } Z$) at various temperatures/frequencies. In view of these difficulties, the “Dielectric Data Plotting and Analyzing Tool (DDPAT)” mainly calculates all possible impedance parameters and sorts the data in the required format for Origin software for plotting and analyzing. Along with this, the tool is also useful in Complex Impedance Spectroscopy(CIS) analysis, especially for Cole–Cole fitting. This tool sorts the data (eg. $\text{Real } Z$, $\text{Img } Z$, Freq) and creates separate text files for all or the selected range of temperatures as per the format of input data required for using the EIS software.



Dielectric Data Plotting and Analyzing Tool [Copyright@Abhinav K., IIT Hyderabad]

Parameters informations

Paste input text file location:

Thickness(mm):

Diameter(mm):

Sample Name:

You want all output files? ☐ Yes

Three axes output and cole-cole files

Do you need this? ☐ Yes

X axis (Freq/Temp): Temp(oC)

Y axis: Real Z(ohm-m)

Z axis: Img Z(ohm-m)

Initial points(colomn):

Final points(colomn):

Initial points(row):

Final points(row):

Select X, Y and Z axes for plot

Do you want plotting? ☐ Yes

Freq/Temp curve: Freq(Hz)

X axis: Temp(oC)

Y axis: Real E

Type of X axis scale: linear

Type of Y axis scale: linear

Initial points of range:

Final points of range:

☐ Given informations are okay

Run the program

Fig 1(1). Graphical user interface(GUI) of “Dielectric Data Plotting and Analyzing Tool”

Procedure for using DDPAT

1(A) Installation of DDPAT

- (1) Download and extract the Zip file of “Dielectric Data Plotting and Analyzing Tool” and save it on the desktop.
- (2) Open the application. GUI window (similar to Fig 1(1)) will pop up on the screen.
- (3) Fill in the information parameters displayed on GUI window as per instructions in the next section 1(B).



1(B) Input file-related instructions:

Some basic precautions should be taken care of before running the program which is as follows:

1. The input file should be as .txt file. If it is not then, save the excel input data as “text delimited”.
 2. Fill in the “Paste input text file location”, “Thickness (mm)”, and “Diameter (mm)” in the parameters information box.
- Eg: Paste input text file location: copy the path as following image.

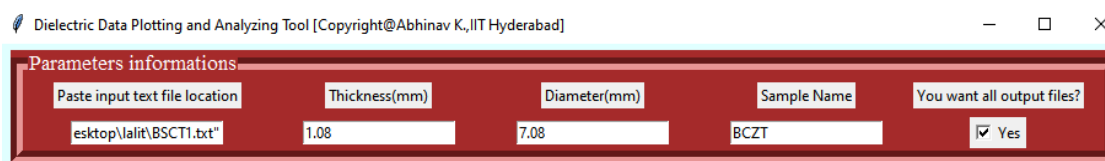
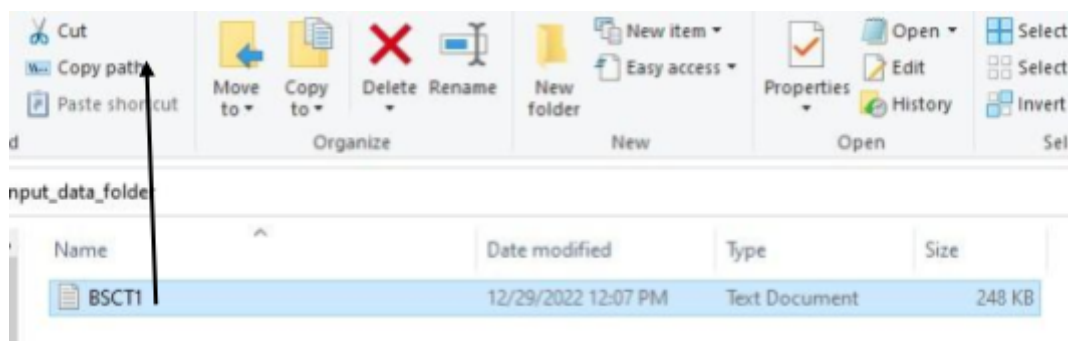


Fig 1(2). Snap of GUI window for parameter information

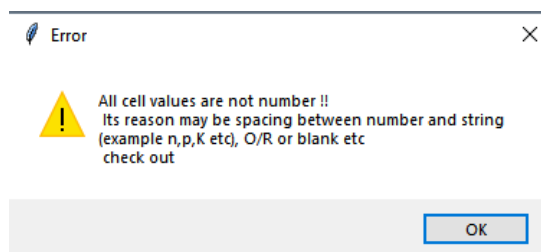


3. Format or order of basic parameters of the input text file should be as follows:

Frequency(Hz)	Temperature(C)	D-Factor(D)	Capacitance(F)	Impedance(Ohm)
20	26	0.039498	1.299850n	6.117275M
20	28	0.035745	1.275665n	6.234129M
20	30	0.031235	1.240651n	6.411076M
20	32	0.027282	1.195090n	6.656203M
20	34	0.025117	1.145928n	6.942167M
20	36	0.024077	1.109503n	7.170248M
20	38	0.020878	1.067113n	7.455254M
20	40	0.022657	1.032267n	7.707115M
20	42	0.024207	0.992359n	8.017070M
20	44	0.021747	0.959394n	8.293573M
20	46	0.022023	0.920796n	8.640296M

Fig 1(3). Snap of data of input text file

4. If in the cell, there is space between the string and number such as 0.3445 n, 7888 M, 99999 K (especially for the Waynekerr analyzer) then remove the spacing by using replacement in the excel.
5. If the cell contains "O/R" value (Check it in excel), then replace the "O/R" with the 1.
6. All cells must contain some valid value i.e. numeric value or numeric value with the string such n, m, M etc.
7. Points 3, 4,5 and 6 must be taken care against following the error message:





8. Suggestions for the smooth running of the tool:

1. Make the new folder and keep the input data text file and copy the input data text location and paste on the GUI window.
 2. Don't run the software if any of the output text files are open.
 3. Don't close the plot for re-running the software as it will automatically re-plot.
 4. If the program is not running, please close the software and re- run the software.
 5. The value of initial points should always be lesser than the value of final points in the selection of the range.
 6. This tool is not for constant frequency/temperature dielectric data.
-



1(C). Output files and plots: all output files are saved in the folder of input data text file (Note: Shown output files and plots are for BCZT material)

1. Output files corresponding to tick “Yes” for “You want all output files?”:

Dielectric Data Plotting and Analyzing Tool [Copyright@Abhinav K., IIT Hyderabad]

Parameters informations

Paste input text file location: Thickness(mm): Diameter(mm): Sample Name: You want all output files?: ☒ Yes

BCZT_Freq_Vs_all_parameters 12/21/2022 10:45 PM File folder

BCZT_Temp_Vs_all_parameters 12/21/2022 10:45 PM File folder

BCZT_Temp(oC) Vs Capacitance(F)	12/21/2022 10:45 PM	Text Document	72 KB
BCZT_Temp(oC) Vs Conductivity(Sm-1)	12/21/2022 10:45 PM	Text Document	147 KB
BCZT_Temp(oC) Vs Freq(Hz)	12/21/2022 10:45 PM	Text Document	55 KB
BCZT_Temp(oC) Vs Img E	12/21/2022 10:45 PM	Text Document	122 KB
BCZT_Temp(oC) Vs Img M	12/21/2022 10:45 PM	Text Document	147 KB
BCZT_Temp(oC) Vs Img Y(ohm-1)	12/21/2022 10:45 PM	Text Document	145 KB
BCZT_Temp(oC) Vs Img Z(ohm-m)	12/21/2022 10:45 PM	Text Document	124 KB
BCZT_Temp(oC) Vs Mod E	12/21/2022 10:45 PM	Text Document	122 KB
BCZT_Temp(oC) Vs Mod M	12/21/2022 10:45 PM	Text Document	144 KB
BCZT_Temp(oC) Vs Mod Y(ohm-1)	12/21/2022 10:45 PM	Text Document	145 KB
BCZT_Temp(oC) Vs Mod Z(ohm)	12/21/2022 10:45 PM	Text Document	77 KB
BCZT_Temp(oC) Vs Mod Z(ohm-m)	12/21/2022 10:45 PM	Text Document	124 KB
BCZT_Temp(oC) Vs Real E	12/21/2022 10:45 PM	Text Document	122 KB
BCZT_Temp(oC) Vs Real M	12/21/2022 10:45 PM	Text Document	145 KB
BCZT_Temp(oC) Vs Real Y(ohm-1)	12/21/2022 10:45 PM	Text Document	148 KB
BCZT_Temp(oC) Vs Real Z(ohm-m)	12/21/2022 10:45 PM	Text Document	123 KB
BCZT_Temp(oC) Vs tan(delta)	12/21/2022 10:45 PM	Text Document	111 KB
BCZT_Temp(oC) Vs Temp(oC)	12/21/2022 10:45 PM	Text Document	44 KB
BCZT_Temp(oC) Vs all parameters	12/21/2022 10:45 PM	Text Document	2.042 KB

Fig 1(4). Snap of GUI window for parameter informations and

output files.



2. Output files corresponding to tick “Yes” for three axes output and Cole-Cole files?”

Three axes output and cole-cole files

Do you need this?	X axis (Freq/Temp)	Y axis	Z axis
☒ Yes	Temp(oC) ▾	Real Z(ohm-m) ▾	Img Z(ohm-m) ▾
0.00	10.00	0.00	10.00
Initial points(column)	Final points(column)	Initial points(row)	Final points(row)
0 2 4 6 8 10	0 2 4 6 8 10	0 2 4 6 8 10	0 2 4 6 8 10

162.0	12/21/2022 10:45 PM	Text Document	4 KB
164.0	12/21/2022 10:45 PM	Text Document	4 KB
166.0	12/21/2022 10:45 PM	Text Document	4 KB
168.0	12/21/2022 10:45 PM	Text Document	4 KB
170.0	12/21/2022 10:45 PM	Text Document	4 KB
172.0	12/21/2022 10:45 PM	Text Document	4 KB
174.0	12/21/2022 10:45 PM	Text Document	4 KB
176.0	12/21/2022 10:45 PM	Text Document	4 KB
178.0	12/21/2022 10:45 PM	Text Document	4 KB
180.0	12/21/2022 10:45 PM	Text Document	4 KB
182.0	12/21/2022 10:45 PM	Text Document	4 KB
184.0	12/21/2022 10:45 PM	Text Document	4 KB
186.0	12/21/2022 10:45 PM	Text Document	4 KB
188.0	12/21/2022 10:45 PM	Text Document	4 KB
190.0	12/21/2022 10:45 PM	Text Document	4 KB
192.0	12/21/2022 10:45 PM	Text Document	4 KB
194.0	12/21/2022 10:45 PM	Text Document	4 KB
196.0	12/21/2022 10:45 PM	Text Document	4 KB
198.0	12/21/2022 10:45 PM	Text Document	4 KB
200.0	12/21/2022 10:45 PM	Text Document	4 KB
BCZT_Freq(Hz) Vs Real Z(ohm-m) Vs Img...	12/21/2022 10:45 PM	Text Document	246 KB

Fig 1(5). Snap of GUI window for three-axis output and Cole-Cole plot files and corresponding output files.



3. Select X, Y and Z axes for the plot:

Select X, Y and Z axes for plot

Do you want plotting? ☒ Yes

Freq/Temp curve: Freq(Hz) [2.92] [8.87]

X axis: Temp(oC) Y axis: Real E

Type of X axis scale: linear Type of Y axis scale: linear

Initial points of range: [0] [2] [4] [6] [8] [10]

Final points of range: [0] [2] [4] [6] [8] [10]

Figure 1

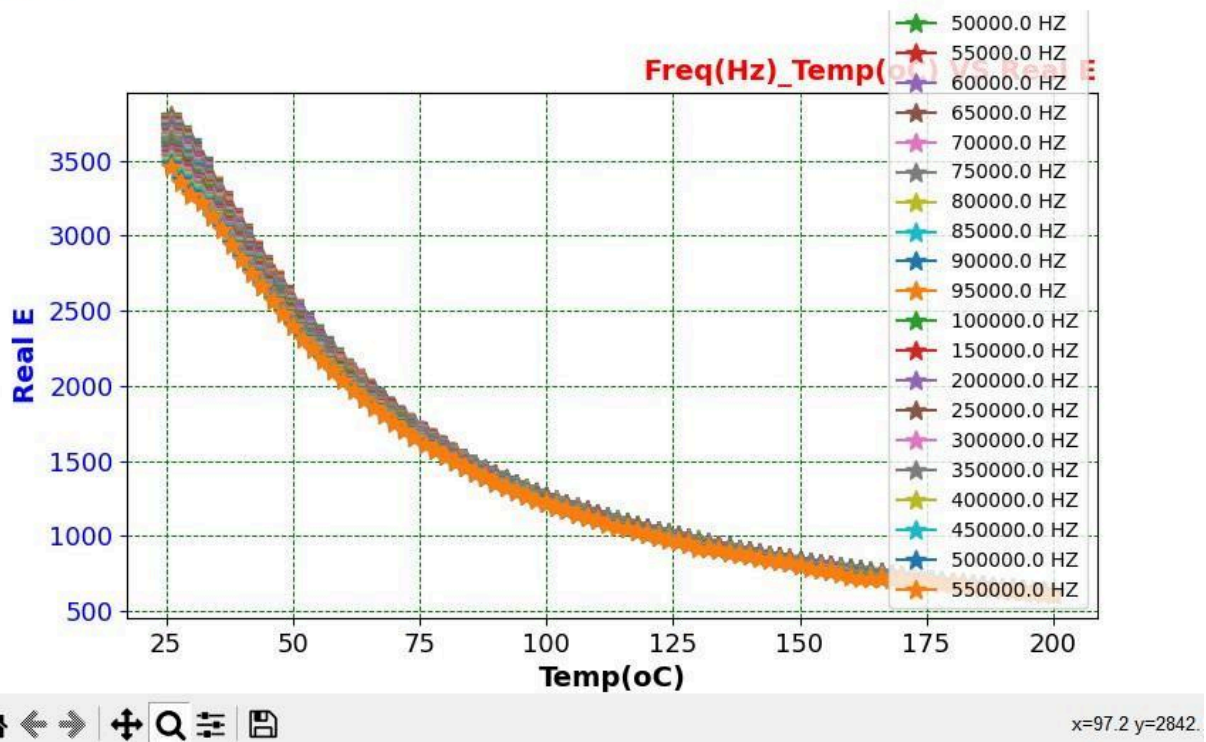


Fig 1(6). Snap of GUI window and plot for temperature-dependent dielectric plot for all or the selected range of frequencies.

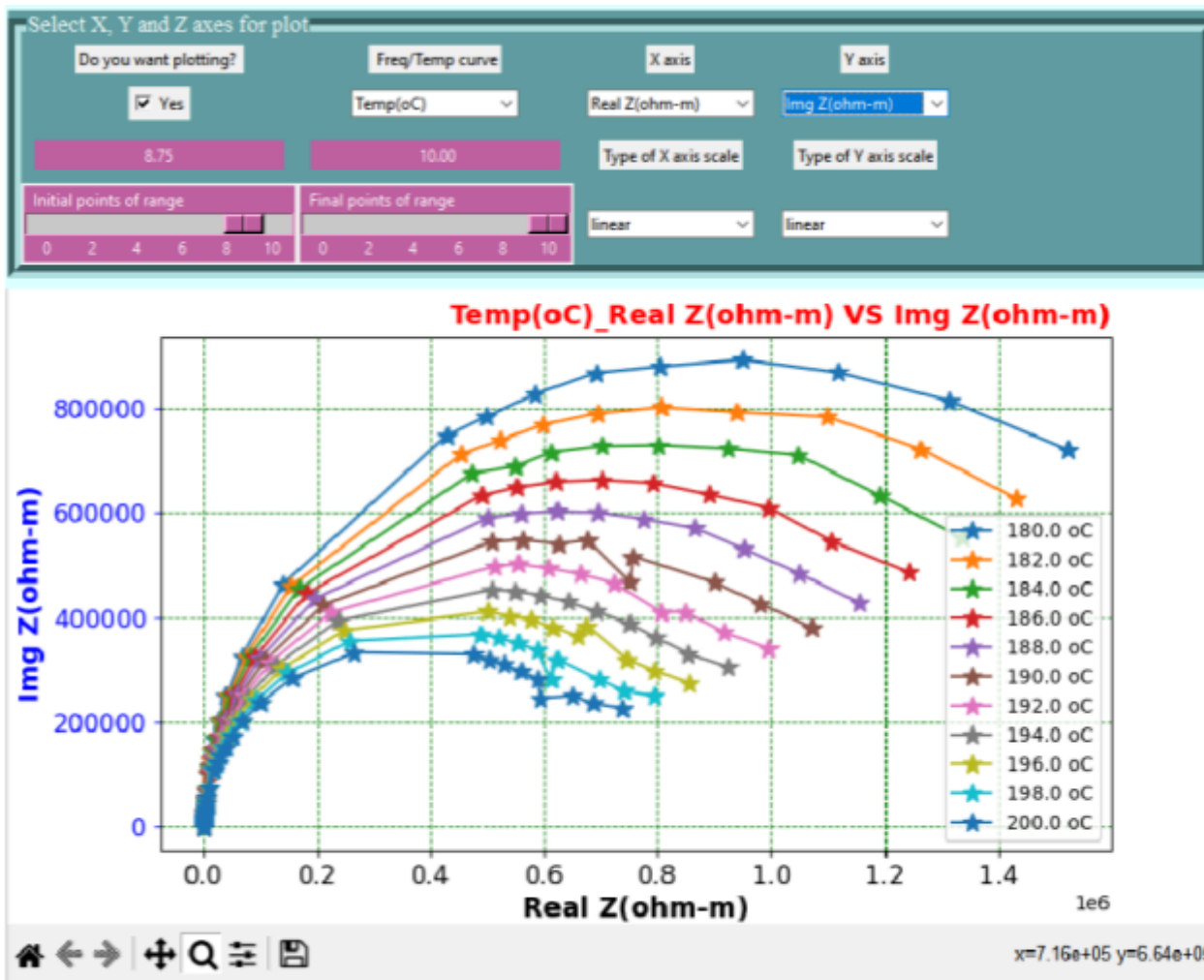


Fig 1(7). Snap of GUI window and plot for Cole-Cole for all or the selected range of temperatures.



Select X, Y and Z axes for plot

Do you want plotting? ☒ Yes

Freq/Temp curve: Temp(oC) [v]

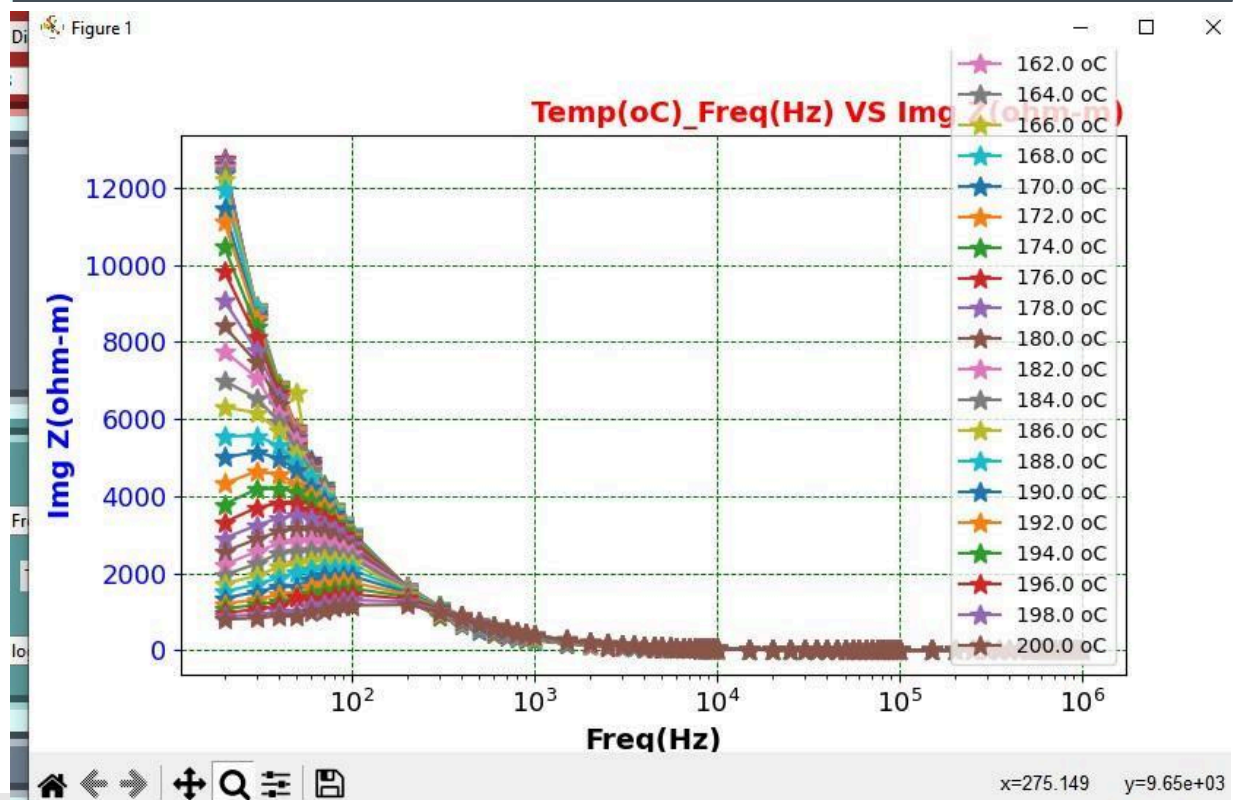
X axis: Freq(Hz) [v] Y axis: Img Z(ohm-m) [v]

5.06 10.00

Type of X axis scale: linear [v] Type of Y axis scale: linear [v]

Initial points of range: 0 2 4 6 8 10

Final points of range: 0 2 4 6 8 10



**Fig 1(8). Snap of GUI window and plot for frequency-dependent
Img Z plot at all or selected range of temperatures.**



Tool [2]

Polarization Data Plotting and Analyzing Tool (PDPAT)

Polarization measured is carried out in category of Dynamic Hysteresis measurement(DHM) and Piezo measurement(PZM) under the wide range of applied electric field or voltage using TF analyzer, aixACCT system. Data in form of table is generated for each applied electric field or voltage. Each table of DHM category is made of parameters (columns in a row)- Time [s], V+ [V], V- [V], I1 [A], P1 [$\mu\text{C}/\text{cm}^2$], I2 [A]", P2 [$\mu\text{C}/\text{cm}^2$] , I3 [A] , P3 [$\mu\text{C}/\text{cm}^2$] , D1 [nm] , D2 [nm] whereas each table under PZM category consists parameters Time [s], V+ [V], V- [V], I1 [A], P1 [$\mu\text{C}/\text{cm}^2$], I2 [A]", P2 [$\mu\text{C}/\text{cm}^2$] , I3 [A] ,P3 [$\mu\text{C}/\text{cm}^2$] , D1 [nm] , D2 [nm]. All the measured tables are saved vertically in a DAT file. In order to analysis and plot the data using Origin software, All tables (not parameters) need to be arranged from vertical to horizontal. Similar to this, in order to estimate energy storage properties and area for scaling behaviors using the Origin software, polarization data of each table need to be arranged in upper curve data and lower curve data, and positive upper curve data and positive lower curve data respectively. "Polarization Data Plotting and Analyzing Tool" addresses the above issues and give the proper formatting of data for analyzing and plotting in origin software.



Polarization Data Plotting and Analyzing Tool [Copyright@Abhinav K., IIT Hyderabad]

Parameter informations

Paste input text file location:

Thickness(mm):

Sample Name:

You want all output files? ☐ Yes

Select X, Y and Z axes for polarization Plot

Do you want plotting? ☐ Yes

Curve: Polarization plot

X axis: Electric Field [kV/cm]

Y axis: P1 [$\mu\text{C}/\text{cm}^2$]

0.00 0.00

Type of X axis scale: linear

Type of Y axis scale: linear

Initial points of range: 0 2 4 6 8 10

Final points of range: 0 2 4 6 8 10

Polarization Parameters Plot

Do you want plotting? ☐ Yes

Polarization parameters: Polarization parameters

X axis: Electric Field [kV/cm]

Y axis: Pmax [$\mu\text{C}/\text{cm}^2$]

☐ Given informations are okay

Run the program

Fig 2.1. Graphical user interface(GUI) of “Dielectric Data Plotting and Analyzing Tool”

Procedure for using PDPAT

2(A) Installation of PDPAT

- (1) Download and extract the Zip file of “Polarization Data Plotting and Analyzing Tool” and save it on the desktop.
- (2) Open the application. GUI window (similar to Fig 2.1) will pop up on the screen.
- (3) Fill in the information parameters displayed on GUI window as per instructions in the next section 2(B).

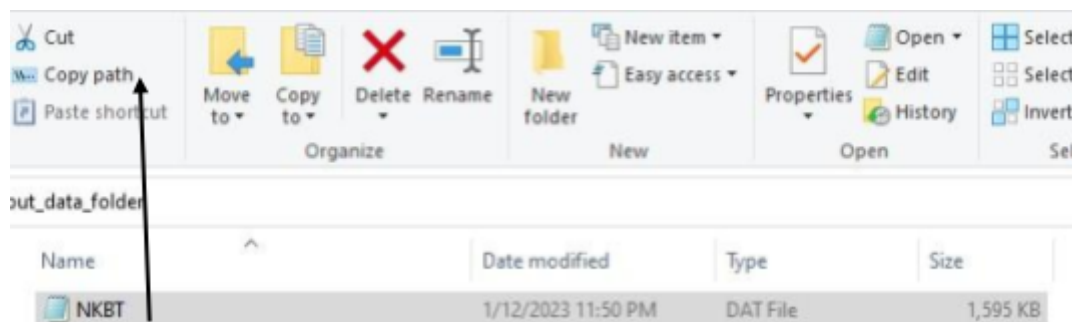


2(B) Input file-related instructions

Some basic precautions should be taken care of before running the program which is as follows:

1. The input file should be as .DAT file (TF analyzer, aixACCT system)
2. Fill in the “Paste input text file location”, “Thickness (mm)” (calculation of electric field parameters from applied voltage voltage), and “Sample Name” in the parameters information box.

Eg: Paste input text file location: copy the path as following image.

A screenshot of a GUI window titled 'Parameter informations'. It has a red border and a white background. There are four input fields: 'Paste input text file location', 'Thickness(mm)', 'Sample Name', and 'You want all output files?'. The first three fields are empty text boxes. The fourth field is a checkbox labeled 'Yes'.

Paste input text file location	Thickness(mm)	Sample Name	You want all output files?
			☐ Yes

Fig 2.2. Snap of GUI window for parameter information



3. Format of the input text file should be as follows:

(1) Input data in DHM category:

```

Wloss [uJ/cm2]: 599231
Cls [F]: 1.35359e-009
Epsls [1]: 3092.27
Basic System: TFAalyzer 2000 E
Measure Head: FE 513-2
MeasureHal Profile: default
Program Version: 3.0.19.0
Operator: Unknown
HV Amplifier: TREK 610E 10kV
Time [s]      V+ [V]  V- [V]  I1 [A]  P1 [uC/cm2]  I2 [A]  P2 [uC/cm2]  I3 [A]  P3 [uC/cm2]  I3 [A]  P3 [uC/cm2]
0.000000e+000  7.098208e+001 -6.861970e+001  2.318538e-005 -2.659285e+001  7.536743e-006 -2.608572e+001 -7.429132e-006  2.559238e+001
2.500000e-003  2.220642e+002 -2.188214e+002  2.405147e-005 -2.645865e+001  9.649491e-006 -2.603690e+001 -9.592483e-006  2.554402e+001
5.000000e-003  3.728404e+002 -3.696583e+002  2.500272e-005 -2.631929e+001  1.068888e-005 -2.597912e+001 -1.071497e-005  2.548633e+001
7.500000e-003  5.231725e+002 -5.201340e+002  2.605265e-005 -2.617425e+001  1.197826e-005 -2.591472e+001 -1.197816e-005  2.542186e+001
1.000000e-002  6.739641e+002 -6.704013e+002  2.769763e-005 -2.602155e+001  1.353898e-005 -2.584223e+001 -1.343487e-005  2.534966e+001
1.250000e-002  8.239843e+002 -8.211413e+002  2.866581e-005 -2.586143e+001  1.512626e-005 -2.576080e+001 -1.520214e-005  2.526831e+001
1.500000e-002  9.739854e+002 -9.709466e+002  3.015927e-005 -2.569431e+001  1.719710e-005 -2.566897e+001 -1.706832e-005  2.517663e+001
1.750000e-002  1.124380e+003 -1.120776e+003  3.157940e-005 -2.551891e+001  1.933560e-005 -2.556518e+001 -1.940540e-005  2.507301e+001
2.000000e-002  1.273664e+003 -1.270586e+003  3.248059e-005 -2.533693e+001  2.143239e-005 -2.544936e+001 -2.183025e-005  2.495586e+001

```

Fig 2.3. Snap of Data of input DAT file(DHM category)

(2) Input data in PHM category:

```

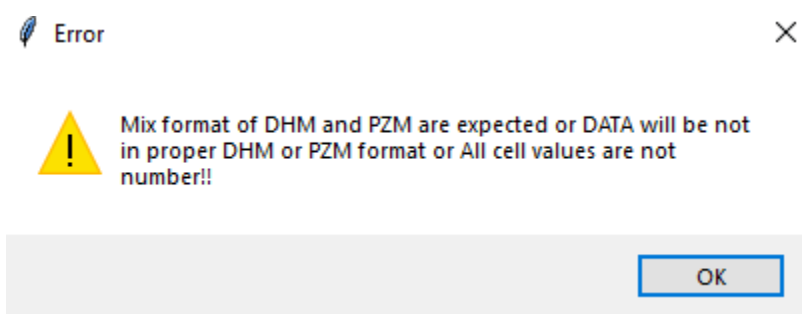
Program Version: 3.0.19.0
Operator: Unknown
HV Amplifier: TREK 610E 10kV
Interferometer: SIOS
CompensateLeakage: 2
DLCC Factor: 0.5
Leakage Compensation Ref Table No: 6
Time [s]      V+ [V]  V- [V]  I1 [A]  P1 [uC/cm2]  I2 [A]  P2 [uC/cm2]  I3 [A]  P3 [uC/cm2]  D1 [nm]  D2 [nm]  D3 [nm]  CH3 [V]
0.000000e+000  9.158723e+000  0.000000e+000  3.429333e-006 -6.126840e-001  3.429333e-006 -6.126840e-001  0.000000e+000 -2.039324e+000 0.0
2.500000e-003  2.455800e+001  0.000000e+000  3.250290e-006 -5.903411e-001  3.250290e-006 -5.903411e-001  0.000000e+000 -2.039324e+000 -1.
5.000000e-003  3.902065e+001  0.000000e+000  2.575017e-006 -5.708559e-001  2.575017e-006 -5.708559e-001  0.000000e+000 -2.039324e+000 -1.
7.500000e-003  5.426755e+001  0.000000e+000  2.807685e-006 -5.528511e-001  2.807685e-006 -5.528511e-001  0.000000e+000 -2.039324e+000 -3.
1.000000e-002  6.915338e+001  0.000000e+000  2.491358e-006 -5.351262e-001  2.491358e-006 -5.351262e-001  0.000000e+000 -2.039324e+000 -9.
1.250000e-002  8.403578e+001  0.000000e+000  2.333913e-006 -5.189860e-001  2.333913e-006 -5.189860e-001  0.000000e+000 -2.039324e+000 -3.
1.500000e-002  9.952324e+001  0.000000e+000  2.674000e-006 -5.022349e-001  2.674000e-006 -5.022349e-001  0.000000e+000 -2.039324e+000 -7.
1.750000e-002  1.144458e+002  0.000000e+000  2.252547e-006 -4.857560e-001  2.252547e-006 -4.857560e-001  0.000000e+000 -2.039324e+000 -1.
2.000000e-002  1.293541e+002  0.000000e+000  2.507654e-006 -4.698334e-001  2.507654e-006 -4.698334e-001  0.000000e+000 -2.039324e+000 -1.
2.250000e-002  1.448029e+002  0.000000e+000  2.656876e-006 -4.525584e-001  2.656876e-006 -4.525584e-001  0.000000e+000 -2.039324e+000 -1.

```

Fig 2.4. Snap of Data of input DAT file(PZM category)



4. Data must be either DHM format or PZM format (as shown above) otherwise it will show following the error message:



5. Suggestions for the smooth running of the tool:

- (1) Make the new folder and keep the input data text file and copy the input data text location and paste on the GUI window.
 - (2) Don't run the software if any of the output text files are open.
 - (3) Don't close the plot for re-running the software as it will automatically re-plot.
 - (4) If the program is not running, please close the software and re-run the software.
 - (5) The value of initial points should always be lesser than the value of final points in the selection of the range.
-



2(C) Output files and plots: all output files are saved in the folder of input data text file ((Note: Shown output files and plots are for NKBT material))

1. Output files corresponding to tick “Yes” for “You want all output files?”:

Parameter informations

Paste input text file location: at_8.8kvHysterFile1.dat

Thickness(mm): 1

Sample Name: NKBT

You want all output files? ☒ Yes

NKBT_Output_P_E_loop_Data_files 1/13/2023 1:01 PM File folder

Fig 2.3. Snap of GUI window for parameter information and output files.

Name	Date modified	Type	Size
NKBT_ES_sorted_data_all_tables	1/13/2023 1:01 PM	Text Document	443 KB
NKBT_Outpule_data_all_tables	1/13/2023 1:01 PM	Text Document	1,669 KB
NKBT_Polarization_parameters	1/13/2023 1:01 PM	Text Document	5 KB
NKBT_Scaling_sorted_data_all_tables	1/13/2023 1:01 PM	Text Document	557 KB

Fig 2.5. Snap of GUI window for output files.



3. Select X, Y and Z axes for the polarization plot:

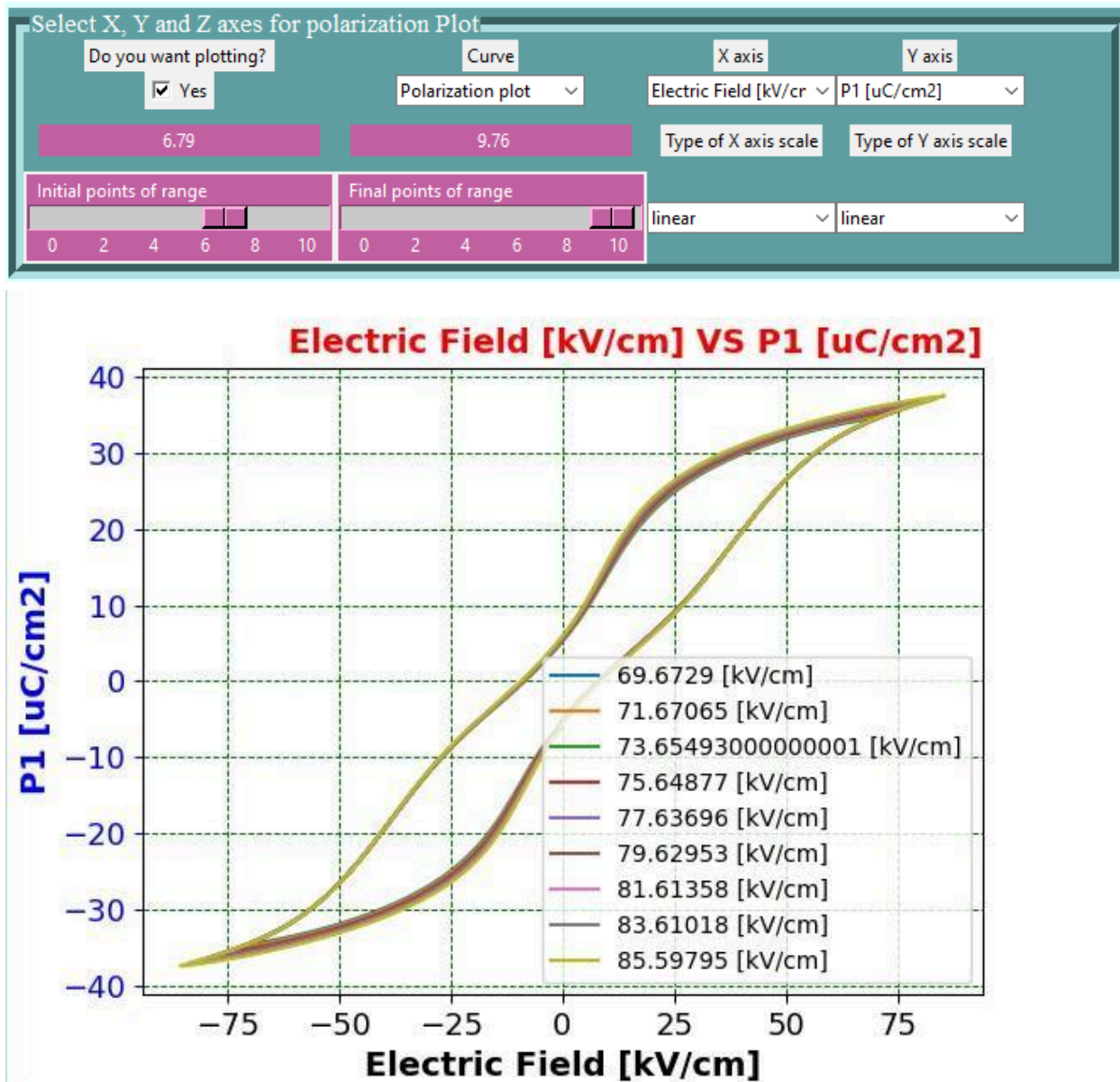


Fig 2.6. Snap of GUI window and plot for electric field vs polarization plot for all or the selected range of applied electric field.



Select X, Y and Z axes for polarization Plot.

Do you want plotting? ☒ Yes

Curve: Polarization plot

X axis: Time[s] Y axis: I1 [A]

6.79 9.76

Type of X axis scale: linear Type of Y axis scale: linear

Initial points of range: 0 2 4 6 8 10

Final points of range: 0 2 4 6 8 10

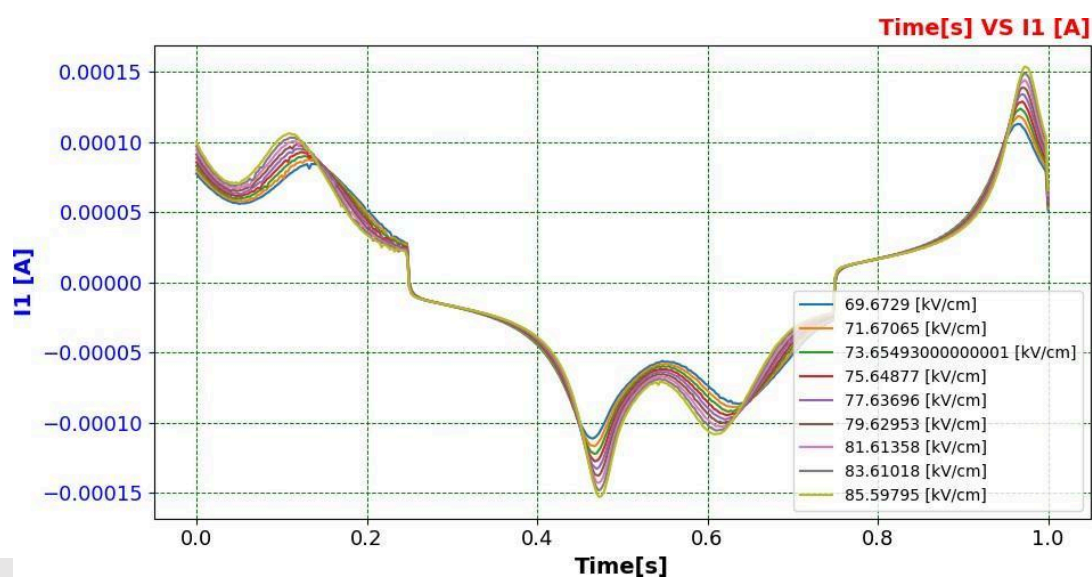


Fig 2.7. Snap of GUI window and plot for current vs time plot for all or the selected range of applied electric field.



Select X, Y and Z axes for polarization Plot

Do you want plotting? ☒ Yes

Curve: Polarization plot

X axis: Electric Field [kV/cm] Y axis: I1 [A]

6.79 9.76

Type of X axis scale: linear Type of Y axis scale: linear

Initial points of range: 0 2 4 6 8 10 Final points of range: 0 2 4 6 8 10

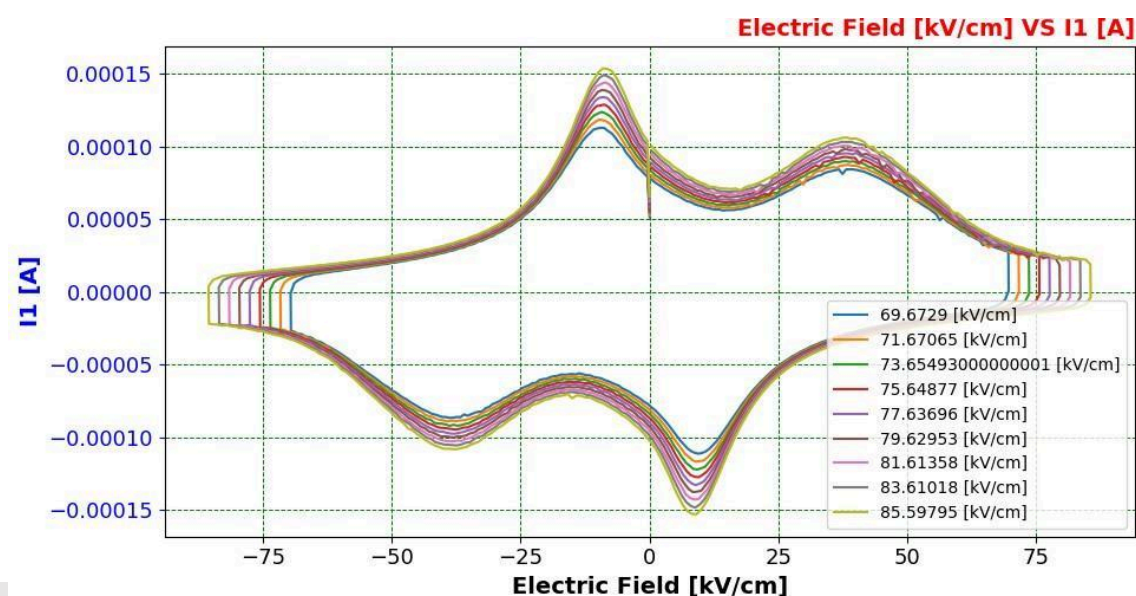


Fig 2.8. Snap of GUI window and plot for current vs electric field plot for all or the selected range of applied electric field



Polarization Parameters Plot

Do you want plotting? ☒ Yes

Polarization parameters

X axis: Electric Field [kV/cm]

Y axis: Pmax [uC/cm2]

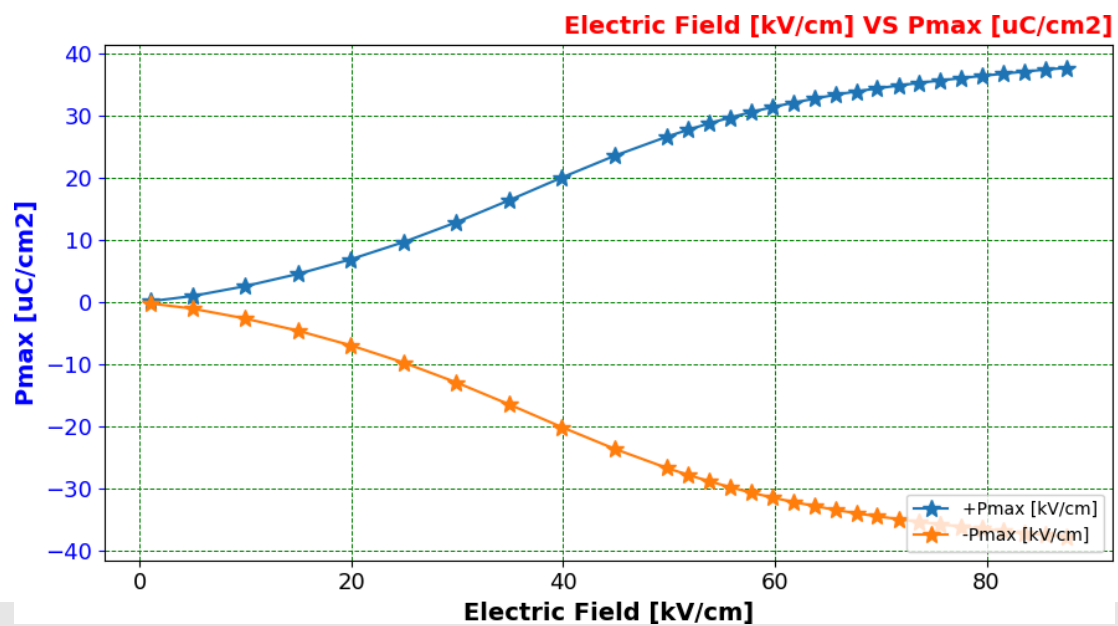


Fig 2.9. Snap of GUI window and plot for maximum polarization Vs electric field plot for all tables(all applied electric fields).



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